Robust Cascade Control of Electric Motor Drives Using Dual Reduced-Order PI Observer

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***Abstract*—This paper addresses the problem of preserving the nominal performance of a conventional cascade controller for electric motor drives in the presence of parameter uncertainties and external disturbances. Practical reduced-order proportional-integral (PI) observers are incorporated with predesigned PI controllers in order to enhance the robust performance of current and speed controllers. A unified analysis based on the singular perturbation theory is presented to confirm the desired approximation of the augmented system with the PI observers to the nominal closed-loop system. The validity of the proposed dual observer-based control scheme is tested via computer simulations and experiments using a dc motor system. Both results verify that the proposed controller can be used effectively to recover the nominal performance of the conventional cascade scheme for electric motor drives.**

***Index Terms*—Cascade control, electric motor drive, proportional–integral (PI) observer, robust control, singular perturbation.**

I. Introduction

OWING to its reliable performance and relatively simple implementation, the proportional–integral–derivative (PID) controller plays a dominant role in a large number of industrial applications [1]. A typical example in this context is cascade control, which is composed of two nested loops with two PID controllers. The controller of the inner loop is the so-called secondary controller, whereas that of the outer loop is the primary controller [2].

One of the most conventional controller designs for electric motor drives consists of proportional-integral (PI) controllers in a cascade configuration [3]. The inner-loop controller regulates the current, and its outer-loop counterpart controls the speed by providing the reference current for the inner loop. Under the presumption of two decoupled first-order systems, the desired closed-loop dynamic specifications can be satisfied by using two simple PI controllers if the system parameters are known precisely. This approach has been applied to the control of dc motors and other types of asynchronous and synchronous machines, as well as power converters [4], [5].

Unfortunately, several electromechanical parameters are not exactly known and/or are subject to large variations during operation, leading to the degradation of the drive performance. Owing to the adverse effects of model uncertainties and external disturbances, the design of efficient controllers has been a central problem in the control society. Several useful solutions, such as adaptive and $\mathcal{H}_\infty$ control, sliding-mode control, model predictive control, and observer-based control, are available in the literature [6]–[18]. These methods have improved the robust performance of the control system in different aspects.

Despite all the advanced control methods available, the conventional cascade configuration can be still adopted as a basis scheme for more demanding control problems because of its practical advantages such as a simple and decoupled controller design for nominal performances. In order to improve the robustness and the disturbance attenuation ability while maintaining the benefits of the conventional cascade scheme and coping with unknown load torque disturbances, disturbance observer-based (DOB) schemes have been considered [12], [26], [27]. However, a single load torque DOB scheme cannot compensate for all electrical and mechanical perturbations, and it is then often necessary to modify the cascade control configuration. Moreover, the load torque observer in previous studies has been utilized without the rigorous analysis of the robustness property against parameter uncertainties.

This paper presents a new practical control structure for the robust control of electric motor drives that preserves both the structure and the nominal performance of the conventional PI-type cascade controller. The proposed structure includes two simple reduced-order PI observers (PIOs) in a cascade structure. Additional feedforward compensation using the dual observers enhances robust performance against parameter uncertainties as well as the slowly varying load torque. In fact, the novelty of the proposed method is not in the use of PIO (or DOB) but in the cascade use of the reduced-order PIOs and the theoretical analysis of the structure. General properties of PIOs are discussed in [20]–[25].

In order to provide a quantitative theoretical analysis of the robustness property against parameter uncertainties, this paper presents a Lyapunov function approach to the robust performance analysis of the proposed observer-based control method. Motivated by the theoretical analysis in [19], the key tool used in this study is the singular perturbation theory [28], which arises from the fast observer dynamics. It will be shown in the paper that the augmented system on the slow manifold (i.e.,

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the quasi-steady-state system) becomes exactly the nominal plant without disturbances. This means that an approximation of the nominal system can be obtained in practice, and the approximation becomes more accurate as the observer poles are placed sufficiently left of the complex plane. Based on the theoretical result, a unified analysis is provided in the state space on the robustness of the proposed cascade scheme.

The proposed controller design process can be divided into two stages. In stage one, nominal cascade PI controllers are designed using a technical optimum scheme [4] to achieve the control objectives in the absence of uncertainties. In stage two, the reduced-order PIOs are constructed to preserve the nominal performance of the cascade control approach. Since the design processes of the dual observers are decoupled from each other, the PIO for the current control loop does not use any information from the mechanical part of the system. The observer for the speed control loop uses neither electrical parameters nor the unknown load torque disturbance.

Computer simulations and experimental studies have been conducted to test the effectiveness of the proposed method. For the purpose of comparison, three different control schemes are considered: (i) the conventional approach with back-electromotive force (EMF) compensation; (ii) a method that uses a single reduced-order PIO with back-EMF compensation; and (iii) a method that uses dual reduced-order PIOs without back-EMF compensation. The results show enhanced performance of the proposed scheme when the third control scheme with the dual reduced-order PIOs is applied.

The main feature of the proposed scheme is that it can cope with parameter uncertainties as well as disturbances such as unmodeled dynamics by using simple first-order dual observers. While the advanced control schemes, e.g., the $\mathcal{H}_\infty$ approach, are in practice restricted by their complex configurations and multiple tuning parameters, the proposed scheme can be simply implemented in practical applications, e.g., electric power steering systems [29], [30], owing to its low dynamic order and robustness property.

The contribution of this paper can be summarized as follows.

1) A new control structure using the dual reduced-order PIOs is presented to preserve both the structure and the nominal performance of the conventional cascade controller for electric motor drives.
2) A Lyapunov function approach is presented as a quantitative theoretical analysis of the robustness property of the proposed PIO-based control method against parameter uncertainties as well as slowly varying disturbances.
3) Both computer simulation and experiment are performed with a dc motor system to demonstrate the effectiveness of the proposed scheme.

This paper is organized as follows. Section II presents the system model and the conventional cascade control scheme along with its performance limits. Section III introduces a reduced-order PIO and analyzes the robust performance of an augmented system combined with the reduced-order PIO. A robustness analysis using a Lyapunov function approach is also presented in the section. Section IV demonstrates the proposed current and speed control schemes with dual reduced-order PIOs. Section V compares the results of the simulation and the experimental tests. Bode plots of the closed-loop systems with three different controllers are also compared. Section VI concludes this paper.

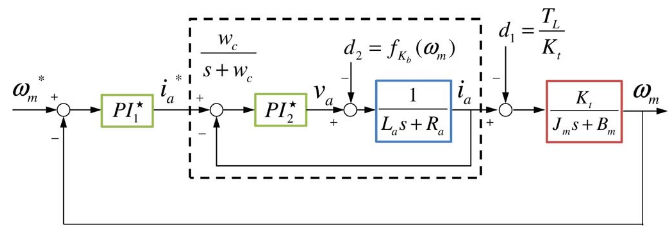

Fig. 1. Cascade control system using ideal controller $PI_1^\star$ of (2) and $PI_2^\star$ of (4) that are designed based on the knowledge of real system parameter and disturbances $d_1(t)$ and $d_2(t)$.

II. System Model and Conventional Control

A. System Model

This paper deals with the current and speed control of a dc motor represented by

$$L_a \frac{di_a}{dt} = -R_a i_a - f_{K_b}(\omega_m) + v_a \quad (1a)$$

$$J_m \frac{d\omega_m}{dt} = -B_m \omega_m - T_L + \tau \quad (1b)$$

where i_a is the armature current, R_a and L_a denote the armature resistance and inductance, respectively, ω_m is the rotor angular speed, $f_{K_b}(\omega_m)$ is the back EMF that is a continuous function of ω_m (which is often assumed to be linear, e.g., $f_{K_b}(\omega_m) = K_b\omega_m$), v_a is the input voltage, J_m and B_m denote the rotor inertia and the friction coefficient, respectively, T_L denotes the load torque disturbance, and the torque $\tau = K_t i_a$ with the torque constant K_t. Since all industrial processes are subject to modeling errors and parameter variations, we suppose that all parameters in (1) have uncertainties and that $f_{K_b}(\omega_m)$ and T_L are unknown whereas $T_L(t)$ is assumed to be slowly varying. For more details about the model (1), readers are referred to, e.g., [3] or [4, Chap. 2].

B. Conventional Cascade Control: Ideal Case

One of the most conventional control schemes for motor drives has the cascade structure where both the inner- and outer-loop controllers are proportional plus integral. As shown in Fig. 1, the outer-loop controller $(PI_1^\star)$ generates the current reference i_a^* for the inner-loop control. The inner loop manipulates the input voltage v_a to make the current i_a track the reference i_a^*. In particular, we review a PI control design based on the technical optimum scheme [4], under the assumptions that all the parameters are known precisely and both the back EMF $f_{K_b}(\omega_m)$ and the load torque T_L are measured.

With the back-EMF term, the current loop controller is designed as

$$v_a = w_c L_a \left(i_a^* - i_a\right) + w_c R_a \int \left(i_a^* - i_a\right) dt + f_{K_b}(\omega_m) \quad (2)$$

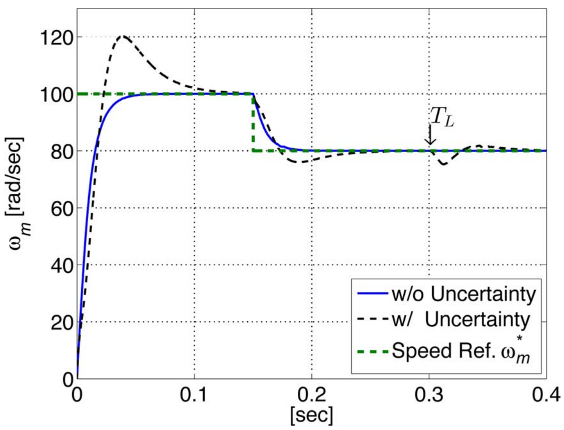

Fig. 2. Simulation result illustrating performance degradation of conventional controller. In the ideal case (blue), real system parameters are equivalent to the nominal ones, while the real parameters in Table II are used in the actual case (black). Speed reference w_m^* is also plotted (green). In both cases, $w_c = 2000$, and $w_s = 100$.

where $w_c > 0$ is a design constant. The term $f_{K_b}(\omega_m)$ cancels the same term in (1a) so that it decouples the effect of ω_m from the dynamics of i_a. Therefore, the transfer function from i_a^* to i_a of the closed-loop system (1a) and (2) can be computed, after a stable pole/zero cancellation, by

$$\frac{\mathrm{i}_a(s)}{\mathrm{i}_a^*(s)} = \frac{w_c(sL_a + R_a)}{(s + w_c)(sL_a + R_a)} = \frac{w_c}{s + w_c} \tag{3}$$

where i_a is the Laplace transform of i_a. It should be noted that w_c determines the bandwidth of the current loop (or the inner loop).

Similarly, for the speed control loop (or the outer loop), assume that the current $i_a(t)$ in (1b) is such that $i_a(t) = i_a^*(t)$ where

$$i_a^* = \frac{w_s J_m}{K_t}(\omega_m^* - \omega_m) + \frac{w_s B_m}{K_t}\int(\omega_m^* - \omega_m)\,dt + \frac{T_L}{K_t} \tag{4}$$

in which ω_m^* is the speed reference and w_s is the desired bandwidth of the outer-loop system. Then, the closed-loop transfer function is obtained similarly as before by

$$\frac{\mathrm{w}_m(s)}{\mathrm{w}_m^*(s)} = \frac{w_s(sJ_m + B_m)}{(s + w_s)(sJ_m + B_m)} = \frac{w_s}{s + w_s} \tag{5}$$

where w_m is the Laplace transform of ω_m.

In the cascade control scheme, w_c is chosen much larger than w_s so that $i_a(t) \approx i_a^*(t)$ in view of the variations of $\omega_m^*(t)$ and $\omega_m(t)$ (or that the transfer function (3) is approximately 1 when $s = j\omega$ in the frequency range $0 \le \omega \le w_s$).

C. Conventional Cascade Control: Nonideal Case

The cascade control of the previous section uses the exact knowledge on system parameters, back EMF, and load torque to be implemented. Therefore, it is not difficult to imagine that the control is not robust and the control performance deteriorates under parameter variations. Indeed, Fig. 2 compares the ideal (denoted by "w/o uncertainty") and nonideal cases (denoted by "w/ uncertainty") where the load torque is unknown (and thus not compensated) under parametric uncertainty. (For the simulation, the nominal and real parameters in Table II are used, and an impulsive load torque disturbance is applied at time 0.3.) Performance degradation owing to model uncertainty and disturbance is apparent in the figure. Thus, in order to deal with these practical issues, this paper presents a (reduced-order) PIO-based compensation method along with a theoretical analysis.

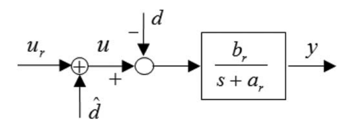

Fig. 3. Real system with compensation using $\hat{d}$.

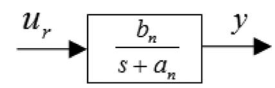

Fig. 4. Nominal system.

TABLE I
MOTOR PARAMETERS FOR OBSERVER DESIGN

	PIO_2 (Current Loop)	PIO_1 (Speed Loop)
a_n (a_r)	$\hat{R}_a/\hat{L}_a$ (R_a/L_a)	$\hat{B}_m/\hat{J}_m$ (B_m/J_m)
b_n (b_r)	$1/\hat{L}_a$ ($1/L_a$)	$\hat{K}_t/\hat{J}_m$ (K_t/J_m)
d	$d_2 = f_{K_b}(\omega_m)$	$d_1 = T_L/K_t$
u	v_a	i_a^*
u_r	u_2	u_1
y	i_a	ω_m

III. Robust Performance via Reduced-Order PIO

A. Introduction to Reduced-Order PIO

In this section, a reduced-order PIO is briefly introduced to explain the proposed robust control with the reduced-order PIO. Since the cascade control approach, reviewed in the previous section, deals with the two first-order systems (1a) and (1b) separately in the same manner, the representative plant of Fig. 3 is studied here which has uncertain parameters a_r and b_r and unknown disturbance d. The goal is to design $\hat{d}$ so that the system in Fig. 3 behaves as if the system in Fig. 4, where a_n and b_n are the nominal values of a_r and b_r, respectively. It will be seen in Section IV that the first-order plant in Fig. 3 corresponds to either (1a) and (1b) when the current or speed loop is considered. In each case, the actual parameters represented by a_r, b_r, and so on are listed in Table I, in which the notation of hat implies the nominal value. The signal u_r can be treated as the reference for now, but it will become the signals u_1 and u_2 that appear in Section IV.

The uncertain system in Fig. 3 is described by

$$\dot{y} = -a_r y + b_r(u - d). \tag{6}$$

Assuming for now that $a_r = a_n$, $b_r = b_n$, and d is constant, we may consider the following system to be a model of (6):

$$\begin{bmatrix}\dot{y}\\ \dot{d}\end{bmatrix} = \begin{bmatrix}-a_n & -b_n\\ 0 & 0\end{bmatrix}\begin{bmatrix}y\\ d\end{bmatrix} + \begin{bmatrix}b_n\\ 0\end{bmatrix}u. \tag{7}$$

Since system (7) is observable w.r.t. the output y, the reduced-order observer can be designed to estimate the disturbance as follows [31]:

$$\dot{\hat{d}} = l(d - \hat{d}) = l\left[\frac{1}{b_n}(-\dot{y} - a_n y + b_n u) - \hat{d}\right] \tag{8}$$

where the observer gain $l > 0$. Then, by the Laplace transformation of both sides, it is obtained that

$$\hat{\mathsf{d}} = \frac{l}{s+l}\mathsf{d}. \tag{9}$$

In order to implement the observer (8) without using the unmeasurable state $\dot{y}$, a new variable ξ is defined as

$$\xi = \hat{d} + \frac{l}{b_n}y, \quad \text{or,} \quad \hat{d} = \xi - \frac{l}{b_n}y. \tag{10}$$

Using (10), the reduced-order observer (8) is rewritten as

$$\dot{\xi} = -l\xi + \frac{l}{b_n}(-a_n + l)y + lu. \tag{11}$$

It is seen from (8) or (9) that $\hat{d}(t)$ converges to the unknown (constant) value d, so the effect of d is successfully attenuated if the estimated $\hat{d}$ is applied as in Fig. 3. Obviously, this holds true only when there is no parameter uncertainty and the disturbance is a constant, i.e., $a_n = a_r$, $b_n = b_r$, and $\dot{d} = 0$. In the next section, we show that the same reduced-order observer can cope with both the parameter uncertainties, i.e., $a_n \neq a_r$ and $b_n \neq b_r$, and the slowly varying disturbance d.

Remark 1: The origin of the name "PIO" becomes clear if the full-order observer for (8) is constructed. Indeed, the full-order observer for (8) is given by

$$\begin{aligned} \dot{\hat{y}} &= -a_n\hat{y} - b_n\hat{d} + b_n u + l_1(y - \hat{y}) \\ \dot{\hat{d}} &= l_2(y - \hat{y}) \end{aligned} \tag{12}$$

which can be equivalently written without the dynamics of $\hat{d}$ as

$$\dot{\hat{y}} = -a_n\hat{y} + b_n u + l_1(y - \hat{y}) - b_n l_2 \int_0^t (y - \hat{y})d\tau \tag{13}$$

because $\hat{d} = l_2 \int_0^t (y - \hat{y})d\tau$. This suggests the name "PIO" while its reduced-order version is used in this paper.

B. Robustness Analysis Using Singular Perturbation Theory

This section shows that the $\hat{d}$ in Fig. 3 that is generated by the PIO can approximately recover the nominal performance of Fig. 4 even in the presence of parameter uncertainties and external disturbance. We make the following assumptions for the system parameters, which are always satisfied with the motor parameters (see Table I).

Assumption 1: The uncertain parameters a_r and b_r belong to known bounded intervals. In addition, the signs of b_n and b_r are the same, and $a_n > 0$ and $a_r > 0$.

To examine the robust performance against parameter uncertainties, the augmented system (6) and (8) is transformed into the standard singular perturbation form [28]. Substituting (6) and $u = u_r + \hat{d}$ into (8) yields

$$\dot{\hat{d}} = \frac{l}{b_n}\left(a_r y - b_r(u_r + \hat{d} - d) - a_n y + b_n u_r\right). \tag{14}$$

When the observer gain l is sufficiently large, the systems (6) and (14) can be written in the standard form of singular perturbation as follows:

$$\dot{y} = -a_r y + b_r(u_r + \hat{d} - d) \tag{15a}$$

$$\epsilon\dot{\hat{d}} = \frac{a_r}{b_n}y - \frac{b_r}{b_n}(u_r + \hat{d} - d) - \frac{a_n}{b_n}y + u_r \tag{15b}$$

where $\epsilon = 1/l$. In the singular perturbation analysis, the variables y, u_r, and d in (15) are referred to as the slow variables, whereas the state $\hat{d}$ is the fast variable.

Under Assumption 1, the boundary-layer system (15b) is stable, and the quasi-steady-state solution $\hat{d}$ of (15b) satisfies

$$-a_r y + b_r(u_r + \hat{d} - d) = -a_n y + b_n u_r. \tag{16}$$

Therefore, in the quasi-steady-state, (15) becomes

$$\dot{y} = -a_n y + b_n u_r \tag{17}$$

which is the same as the nominal system of Fig. 4 without the disturbance and parameter uncertainties. Hence, it can be said that the system of Fig. 3, with (10) and (11), behaves as the nominal system as $l \to \infty$. A more thorough analysis follows.

Proposition 1: Under Assumption 1, the system (15) is stable when $l > 0$. Moreover, if the input u_r and disturbance d are bounded and continuously differentiable with bounded derivatives, then, as l becomes larger, the system (15) behaves more like the nominal system (17) after the fast transient of $\hat{d}(t)$.

Proof: Equation (15) can be rewritten as

$$\begin{cases} \dot{y} = -a_r y + b_r\hat{d} + B_1 v \\ \dot{\hat{d}} = -l\frac{\tilde{a}}{b_n}y - l\frac{b_r}{b_n}\hat{d} + lB_2 v \end{cases} \tag{18}$$

where $B_1 = [b_r \quad -b_r]$, $B_2 = (1/b_n)[\tilde{b} \quad b_r]$, $\tilde{a} = a_n - a_r$, $\tilde{b} = b_n - b_r$, and $v = [u_r \quad d]^T$. The characteristic equation of the linear system (18) is given by

$$\alpha(s) = s^2 + \left(a_r + l\frac{b_r}{b_n}\right)s + l\frac{a_n b_r}{b_n}. \tag{19}$$

Since the two coefficients of (19) are positive under Assumption 1, the system is always stable.

Let $h(y, v) := -(\tilde{a}/b_r)y + (b_n/b_r)B_2 v$ and $e := \hat{d} - h(y, v)$. Then, it can be shown that

$$\begin{cases} \dot{y} = -a_n y + b_n u_r + b_r e \\ \dot{\hat{d}} = -l\frac{b_r}{b_n}e. \end{cases} \tag{20}$$

Note that the y-dynamics in (20) is that of the ideal system (17) when $e = 0$.

In order to compare the behavior of (20) with that of (17), let the solution of (17) be $y_n(t)$. That is, $y_n(t)$ satisfies the equation $\dot{y}_n = -a_n y_n + b_n u_r$. If we define $\tilde{y} := y - y_n$, then

$$\dot{\tilde{y}} = -a_n\tilde{y} + b_r e \tag{21}$$

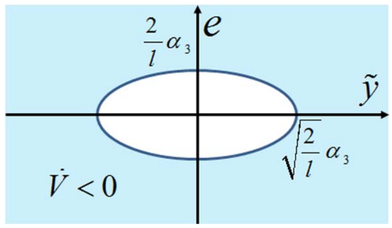

Fig. 5. Region of $\dot{V} < 0$.

and the e-dynamics are written as

$$\dot{e} = -\frac{\tilde{a}a_n}{b_r}\tilde{y} - \left(l\frac{b_r}{b_n} - \tilde{a}\right)e + \bar{B}\theta \tag{22}$$

where $\bar{B} = (1/b_r)[-\tilde{a}a_n \;\; \tilde{a}b_n \;\; -b_nB_2]$ and $\theta = [y_n \;\; u_r \;\; \dot{v}]^T$. Since $u_r(t)$ and $\dot{v}(t) = [\dot{u}_r(t), \dot{d}(t)]^T$ are bounded by the assumption of Proposition 1 and $y_n(t)$ is the solution of the stable system $\dot{y}_n = -a_ny_n + b_nu_r$, it is seen that $\theta(t)$ is bounded.

Let $V(\tilde{y}, e) = (1/2a_n)\tilde{y}^2 + (b_n/2b_r)e^2$. Then, we obtain

$$\dot{V} \le -\tilde{y}^2 - le^2 + \alpha_1 e^2 + \alpha_2|\tilde{y}||e| + \alpha_3|e| \tag{23}$$

where $\alpha_1 = |\tilde{a}b_n/b_r|$, $\alpha_2 = |b_r/a_n| + |\tilde{a}a_nb_n/b_r^2|$, and $\alpha_3 = \|(b_n/b_r)\bar{B}\| \max_{0 \le t \le \infty} \|\theta(t)\|$. Since

$$\alpha_2|\tilde{y}||e| + \alpha_3|e| \le \frac{\tilde{y}^2}{2} + \frac{\alpha_2^2}{2}e^2 + \frac{l}{4}e^2 + \frac{1}{l}\alpha_3^2 \tag{24}$$

by Young's inequality, it follows that

$$\dot{V} \le -\left(\frac{\tilde{y}^2}{2} + \frac{l}{4}e^2 - \frac{1}{l}\alpha_3^2\right) - \left(\frac{l}{2} - \alpha_1 - \frac{\alpha_2^2}{2}\right)e^2. \tag{25}$$

If $l > 2\alpha_1 + \alpha_2^2$, then $\dot{V} < 0$ outside of the ellipse shown in Fig. 5. Since the size of the ellipse can be made arbitrarily small by taking a sufficiently large l, $V(\tilde{y}(t), e(t))$ tends to an arbitrarily small number as time increases, and the error $\tilde{y}(t)$ becomes arbitrarily small. ■

Remark 2: In order to verify the steady-state performance in the frequency domain, the transfer functions of (15) from both u_r and d to y are computed. The transfer function $G_{yu_r}(s)$ from u_r to y is given by

$$G_{yu_r}(s) = \frac{b_r(s+l)}{s\left[s + \left(a_r + l\frac{b_r}{b_n}\right)\right] + l\frac{a_nb_r}{b_n}} \tag{26}$$

whereas $G_{yd}(s)$ from d to y is given by

$$G_{yd}(s) = \frac{-b_rs}{s\left[s + \left(a_r + l\frac{b_r}{b_n}\right)\right] + l\frac{a_nb_r}{b_n}}. \tag{27}$$

It can be seen that the transfer functions are always stable with the motor parameters when $l > 0$. Note that the dc gain of (26) is $G_{yu_r}(0) = b_n/a_n$, which is the same as that of the nominal transfer function, and since $G_{yd}(0) = 0$, the steady-state response to the constant disturbance yields zero.

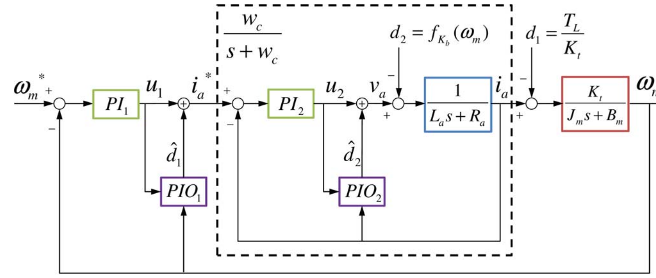

Fig. 6. Proposed cascade control with $\hat{d}_1$ and $\hat{d}_2$ being provided by two reduced-order PIOs. Two PI controllers PI_1 and PI_2 are designed with the nominal parameters as in the ideal case of Fig. 1 but do not have the compensating d_1 and d_2. See (29) and (41).

IV. Cascade Control via Nested PIOs

To maintain the advantages of the conventional cascade control scheme, this study employs a control structure composed of two cascaded PI controllers PI_1 and PI_2. In addition to the cascade PI controllers, two reduced-order PIOs are installed to the current and speed control loops in a cascading manner for robust control.

Fig. 6 shows the proposed control structure, in which the control counteracts the performance degradation owing to parameter uncertainties and external disturbances. The estimates $\hat{d}_1$ and $\hat{d}_2$ in Fig. 6 are obtained from two first-order observers PIO_1 and PIO_2, respectively. Table I lists the system parameters that correspond to the reduced-order observer (11).

For the robust cascade speed control, the current control system is first rendered to approximate (3) under parameter uncertainties by using u_2 and $\hat{d}_2$ in Fig. 6.

A. Current Control System With PIO_2

Since the mechanical response is rather slow compared to the current transients, the back-EMF term $f_{K_b}(\omega_m)$ can be regarded as a slowly varying unknown signal, denoted by d_2 in Fig. 6. Then, the current dynamics can be rewritten as

$$\frac{di_a}{dt} = -\frac{R_a}{L_a}i_a + \frac{1}{L_a}(u_2 + \hat{d}_2 - d_2). \tag{28}$$

With the reference i_a^*, the controller PI_2 is designed as

$$\begin{aligned} \dot{\eta}_2 &= i_a^* - i_a, \\ u_2 &= w_c\hat{L}_a\left(i_a^* - i_a\right) + w_c\hat{R}_a\eta_2. \end{aligned} \tag{29}$$

Note that the nominal parameters are used without the back-EMF compensation. Then, the reduced-order PIO, PIO_2, for the current control loop is designed as in (10) and (11) with $u = u_r + \hat{d}$ by

$$\dot{\xi}_2 = -l_2\hat{R}_ai_a + l_2u_2, \tag{30}$$

$$\hat{d}_2 = \xi_2 - l_2\hat{L}_ai_a \tag{31}$$

where $l_2 > 0$ is the observer gain. (Indeed, (30) follows from (11) with $u = u_r + \hat{d}$ since $\dot{\xi} = -l(\xi - \hat{d} - (l/b_n)y) - l(a_n/b_n)y + lu_r = -l(a_n/b_n)y + lu_r$.)

With Proposition 1 applied, it follows that, in spite of the presence of parameter uncertainties, the system (28) and (30) is stable and can be rendered with (31) to behave like the nominal system

$$\frac{di_a}{dt} = -\frac{\hat{R}_a}{\hat{L}_a} i_a + \frac{1}{\hat{L}_a} u_2 \tag{32}$$

after the transient of fast dynamics, when the observer gain l_2 is sufficiently large. That is, the transfer function from u_2 to i_a, in Fig. 6, can be approximated by

$$\mathsf{i}_a(s) = \frac{1}{\hat{L}_a s + \hat{R}_a} \mathsf{u}_2(s). \tag{33}$$

Consequently, the nominal controller PI_2 could be applied, and the closed-loop system (28)–(31) is approximated by

$$\mathsf{i}_a(s) = \frac{w_c}{s + w_c} \mathsf{i}_a^*(s). \tag{34}$$

In the cascade speed control scheme, the reference i_a^* is generated by the outer-loop speed controller (PI_1) and the estimate $\hat{d}_1$ as shown in Fig. 6.

B. Speed Control System With PIO_1

Assuming that the closed-loop system (28)–(31) is approximated by (34), the cascade connection of (1b) and (34) is represented by (see Fig. 6)

$$\begin{cases} \frac{d}{dt}\omega_m = -\frac{B_m}{J_m}\omega_m + \frac{K_t}{J_m}(i_a - d_1) \\ \frac{1}{w_c}\frac{d}{dt} i_a = -i_a + i_a^* \end{cases} \tag{35}$$

where $d_1 = T_L/K_t$.

Note that the above equation is in the standard singular perturbation form when the current loop bandwidth w_c is sufficiently large. Since the boundary-layer system of (35) is stable, the system can be approximated to the quasi-steady-state system by the singular perturbation theory. This quasi-steady-state system is given by

$$\frac{d}{dt}\omega_m = -\frac{B_m}{J_m}\omega_m + \frac{K_t}{J_m}(i_a^* - d_1) \tag{36}$$

$$= -\frac{B_m}{J_m}\omega_m + \frac{K_t}{J_m}(u_1 + \hat{d}_1 - d_1). \tag{37}$$

Since the load torque varies slowly in practice, the reduced-order observer for the speed control system can be designed as

$$\dot{\xi}_1 = -l_1 \frac{\hat{B}_m}{\hat{K}_t}\omega_m + l_1 u_1, \tag{38}$$

$$\hat{d}_1 = \xi_1 - l_1 \frac{\hat{J}_m}{\hat{K}_t}\omega_m \tag{39}$$

where $l_1 > 0$ is the observer gain. Using Proposition 1, the transfer function from u_1 to ω_m in Fig. 6 [or (37)–(39)] can be approximated by

$$\mathsf{w}_m(s) = \frac{\hat{K}_t}{\hat{J}_m s + \hat{B}_m} \mathsf{u}_1(s) \tag{40}$$

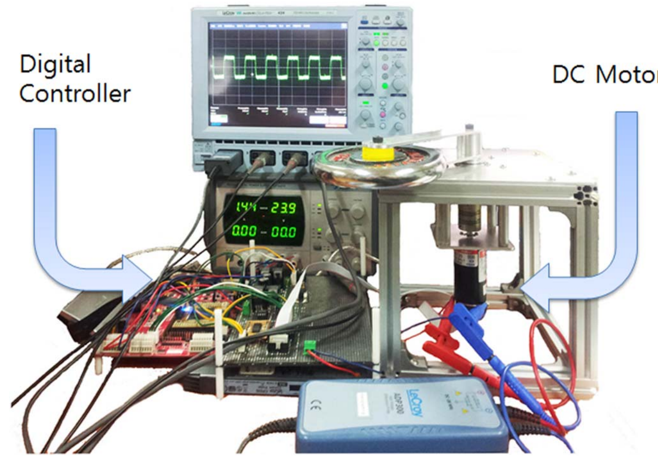

Fig. 7. DC motor system for experiments.

with the estimate (39) using (38). From this, we can design the PI controller

$$\begin{aligned} \dot{\eta}_1 &= \omega_m^* - \omega_m \\ u_1 &= \frac{w_s \hat{J}_m}{\hat{K}_t}(\omega_m^* - \omega_m) + \frac{w_s \hat{B}_m}{\hat{K}_t}\eta_1 \end{aligned} \tag{41}$$

where $\hat{K}_t$, $\hat{J}_m$, and $\hat{B}_m$ denote the nominal values of the system parameters. Note that the term for load torque compensation is absent.

Consequently, the final closed-loop transfer function is approximated by

$$\mathsf{w}_m(s) = \frac{w_s}{s + w_s} \mathsf{w}_m^*(s). \tag{42}$$

This shows that the control objective of the study is accomplished by using the proposed observer-based cascade control scheme, i.e.,

$$\begin{aligned} v_a = &-w_c\hat{L}_a(l_1 + w_s)\frac{\hat{J}_m}{\hat{K}_t}\omega_m - (l_2 + w_c)\hat{L}_a i_a \\ &+ w_c\hat{L}_a w_s\frac{\hat{B}_m}{\hat{K}_t}\eta_1 + w_c\hat{L}_a\xi_1 \\ &+ w_c\hat{R}_a\eta_2 + \xi_2 + w_c\hat{L}_a w_s\frac{\hat{J}_m}{\hat{K}_t}\omega_m^*. \end{aligned} \tag{43}$$

From the discussions so far, it is suggested to set $l_2 \gg w_c \gg l_1 \gg w_s$. Indeed, $l_2 \gg w_c$ ensures that (28)–(31) are approximated by (33), and $w_c \gg l_1$ ensures that (35) becomes (36). Similarly, (37)–(39) are approximated by (40) with $l_1 \gg w_s$.

V. Experiments

This section discusses the testing of the proposed controller through computer simulations and laboratory experiments using the one-link manipulator shown in Fig. 7. The system consists of a dc motor (Maxon RE35), a harmonic drive whose gear ratio is 100, and a rigid link. The parameters are listed in Table II. The uncertain real parameters are specified for the computer simulations.

TABLE II
SYSTEM PARAMETERS FOR EXPERIMENTS

Parameter	Nominal	(Real)	Unit
$\hat{R}_a$ (R_a)	0.1	(0.605)	[Ω]
$\hat{L}_a$ (L_a)	0.076	(0.210)	[mH]
$\hat{K}_t$ (K_t)	0.0292	(0.0234)	[Nm/A]
$\hat{J}_m$ (J_m)	78.70	(86.57)	[g-cm^2]
$\hat{B}_m$ (B_m)	2.1084e-1	(4.2167e-2)	[mNm/(rad/s)]
$\hat{K}_b$ (K_b)	0.0291	(0.0233)	[V/(rad/s)]

TABLE III
THREE DIFFERENT CONTROLLERS

	PI_1	PI_2	Back-EMF	PIO_1	PIO_2
(i) Conventional	O	O	O	X	X
(ii) PIO1:O, PIO2:X	O	O	O	O	X
(iii) PIO1:O, PIO2:O	O	O	X	O	O

Both the simulation result and the experimental result compare the performance of three different control schemes: (i) the conventional approach defined in (2) and (4) with back-EMF compensation; (ii) the method using a single observer PIO_1 with back-EMF compensation; and (iii) the proposed method without back-EMF compensation. Note that all the controllers use the nominal values of parameters and the load torque information remains unknown. For convenience, the three controllers are listed in Table III.

For the experiments, the proposed control algorithms are implemented by using the TI DSP TMS320F28335. All the controllers and observers are discretized through the trapezoidal (or Tustin) method. The sampling period for the current control, including A/D conversion, is 25 μs, which is also the pulse-width modulation period. The speed controller and observer are updated every 100 μs. For the speed information, a position encoder-based estimation is performed using a typical speed observer [4].

The desired current loop bandwidth for PI_2 in (29) is set as $w_c = 2000$, and the observer gain for PIO_2 is chosen as $l_2 = 10\,000$. The speed observer poles are located at $s = -500$. The desired speed loop bandwidth is $w_s = 100$, with which the PI controller (41) is designed. The gain of PIO_1 is given by $l_1 = 500$.

A. Simulation Results

The simulation results are illustrated in Figs. 8 and 9. The speed reference signal is changed from $\omega_m^* = 100$ rad/s to $\omega_m^* = 80$ rad/s at 0.15 s. To verify the robust performance against parameter uncertainties, the nominal values are quite different from the real ones in Table II for simulations. In addition to the parameter uncertainties in Table II, a fast changing disturbance is enforced at 0.3 s as shown in Fig. 8.

Fig. 9 shows the dynamic current control responses during the speed control. The reference signal (i_a^*) is generated by the outer-loop speed controller.

The simulation results show that the proposed method preserves the nominal performance when the dual observers are combined with the PI controllers. In Fig. 8, PIO_2 seems to play an important role when there is a large variation in the reference signal (ω_m^*). By using the dual PIOs, a desired i_a^* has been obtained in Fig. 9(a), and the current control error has been reduced as shown in Fig. 9(b).

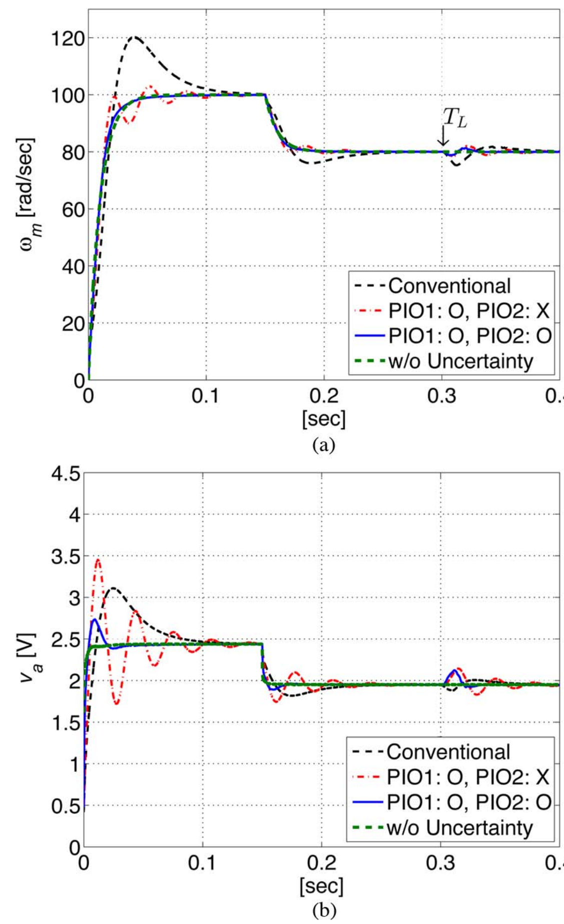

Fig. 8. Simulation results of speed loop. (a) Motor speed ω_m [rad/s]. (b) Input voltage v_a [V].

B. Experimental Results

Fig. 10 shows the experimental results of the speed control. The current control responses during the speed control are illustrated in Fig. 11. Both results show that, unlike methods (i) and (ii), method (iii) maintains the desired performance against parameter uncertainties as well as an external disturbance.

Results of the simulation and the experiment confirm that the proposed algorithm achieves the control objective while preserving the transient response without any steady-state errors. Moreover, it can be seen that the values of i_a^* and v_a of the experiments are larger than those of the simulations. The differences suggest the existence of unmodeled dynamics of the experimental system. This implies that the proposed approach using the dual observers PIO_1 and PIO_2 can successfully cope with parameter uncertainties and unmodeled dynamics, e.g., dynamic friction [17]. Friction force often degrades the performance of mechanical systems at low speed.

It may be possible to improve the performance by increasing the gain of the single observer PIO_1. However, the high-gain approach using a single observer often causes undesirable noise and vibrations.

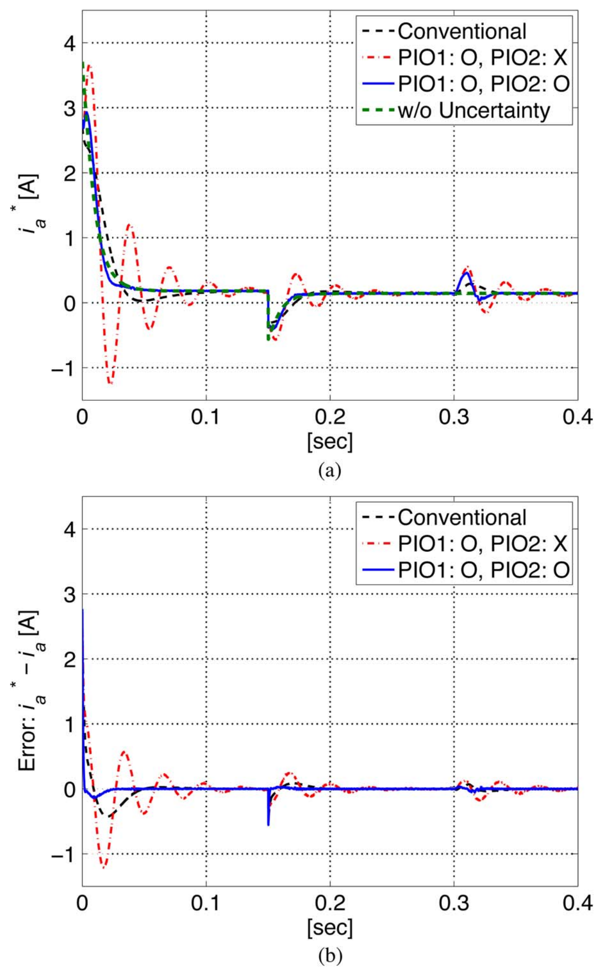

Fig. 9. Simulation results of current loop. (a) Reference current i_a^* [A]. (b) Current control error $i_a^* - i_a$ [A].

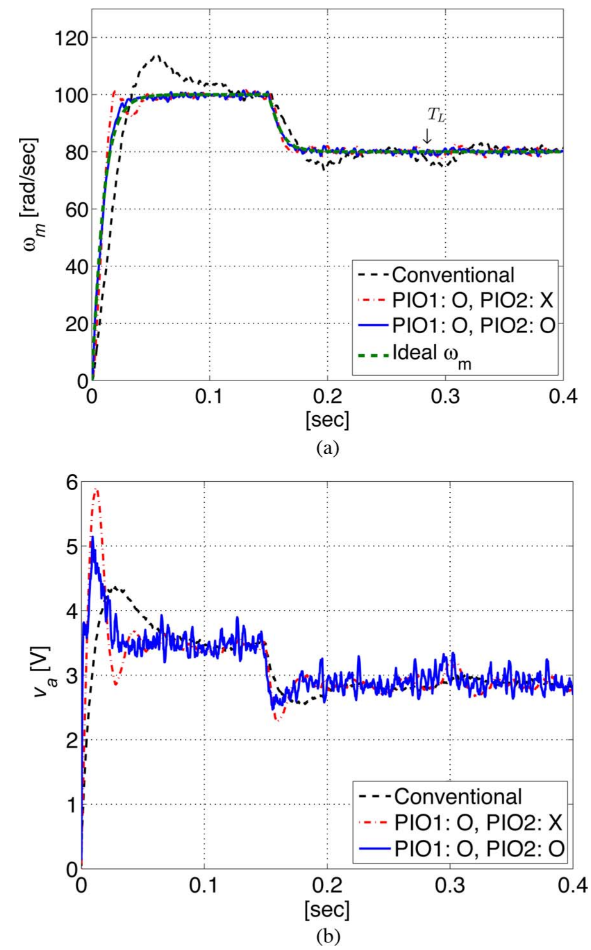

Fig. 10. Experiment results of speed loop. (a) Motor speed ω_m [rad/s]. (b) Input voltage v_a [V].

C. Bode Plots of Cascade Control Systems

In order to study the role of the proposed observers PIO_1 and PIO_2 in the frequency domain, Bode plots of the transfer function G_ω from u_1 to ω_m for the three different methods are compared in Fig. 12. The transfer functions have been computed from the state-space equations in Appendix A.

The two transfer functions using the proposed PIO have the same dc gain as the nominal system. The augmented system using dual observers shows more desirable characteristics than the other systems.

In order to achieve the nominal cascade control performance, the inner loop is needed to maintain the desired bandwidth under parameter uncertainties. Bode plots of the transfer function G_i from i_a^* to i_a can be used to test the robustness of the inner-loop dynamic characteristics.

Since PIO_1 is not included in G_i (see Fig. 6), two cases are compared in Fig. 13. The transfer functions have been obtained from the state-space equations in Appendix B. The figure shows the role of PIO_2 to maintain the desired bandwidth w_c of the nominal system.

It should be mentioned that, when the parameter uncertainties become large, both observer gains l_1 and l_2 should be increased simultaneously in order to achieve the desired performance.

VI. Conclusion

Since the performance of the conventional cascade control method for electric motor drives can easily deteriorate owing to parameter uncertainties and external disturbances, a new practical control structure using dual first-order observers has been proposed. This scheme enables the recovery of the nominal performance of the PI-type cascade controller. While the single full-order PIO-based load torque observer has been widely employed without explicit analysis of the robust performance against parameter uncertainties, this paper presents theoretical analyses of the robust performance by using the singular perturbation theory. A rigorous theoretical analysis using the Lyapunov function has shown that a real uncertain system can be forced to behave like a nominal one when the observer gain is sufficiently large. Moreover, transfer functions of the augmented system show the robust properties of the proposed approach in the steady state. The cascade approach using the proposed observer has been subsequently applied to current and speed control in order to preserve the nominal performance without any disturbance or parametric uncertainties. To verify the performance of the proposed approach, computer simulations and experimental tests have been performed for the robust speed control problem. All the results show that the proposed observer-based scheme can be effectively employed to maintain nominal performance against parameter uncertainties.

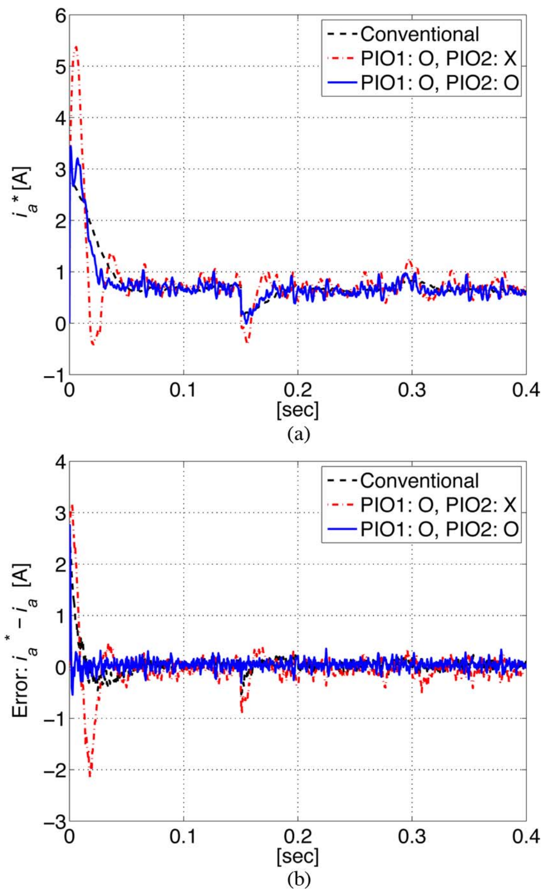

Fig. 11. Experiment results of current loop. (a) Reference current i_a^* [A]. (b) Current control error $i_a^* - i_a$ [A].

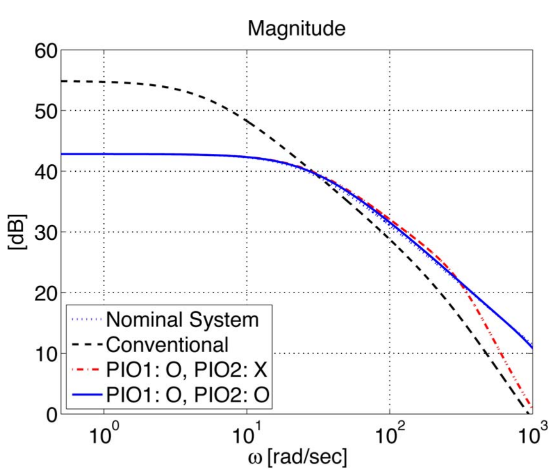

Fig. 12. Bode plots of methods (i), (ii), and (iii).

Moreover, the experimental results confirm that the proposed scheme using the dual observers can improve the robustness against unmodeled dynamics as well as parameter uncertainties. Future study includes an explicit theoretical analysis of the single PIO-based cascade controller and its comparison with the proposed method.

Appendix A
Transfer Functions G_ω

The transfer functions G_ω from u_1 to ω_m in Fig. 6 using the three different control methods (i), (ii), and (iii) in Section V

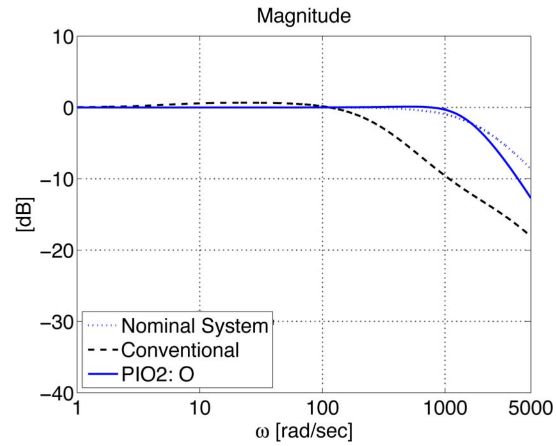

Fig. 13. Bode plots of current control loop.

TABLE IV
System Parameters for G_ω

	x	A	B	C
(i) Conventional	x_c	A_c	B_c	C_c
(ii) PIO1:O, PIO2:X	x_s	A_s	B_s	C_s
(iii) PIO1:O, PIO2:O	x_d	A_d	B_d	C_d

can be obtained by the following state-space equations:

$$\begin{aligned}\dot{x} &= Ax + Bu\\ y &= Cx\end{aligned} \tag{44}$$

where the state x and the matrices are different for the three controllers. The states and matrices are summarized in Table IV.

The system parameters in Table IV are given by the following matrices:

$$x_c = \begin{bmatrix}\omega_m & i_a & \eta_2\end{bmatrix}^T \tag{45}$$

$$A_c = \begin{bmatrix} -\frac{B_m}{J_m} & \frac{K_t}{J_m} & 0\\ \frac{-K_b+\hat{K}_b}{L_a} & \frac{-R_a-w_c\hat{L}_a}{L_a} & \frac{w_c\hat{R}_a}{L_a}\\ 0 & -1 & 0\end{bmatrix} \tag{46}$$

$$B_c = \begin{bmatrix}0 & \frac{w_c\hat{L}_a}{L_a} & 1\end{bmatrix}^T, \quad C_c = \begin{bmatrix}1 & 0 & 0\end{bmatrix} \tag{47}$$

$$x_s = \begin{bmatrix}\omega_m & i_a & \xi_1 & \eta_2\end{bmatrix}^T \tag{48}$$

$$A_s = \begin{bmatrix} -\frac{B_m}{J_m} & \frac{K_t}{J_m} & 0 & 0\\ A_s^{21} & \frac{-R_a-w_c\hat{L}_a}{L_a} & \frac{w_c\hat{L}_a}{L_a} & \frac{w_c\hat{R}_a}{L_a}\\ -\frac{l_1\hat{B}_m}{\hat{K}_t} & 0 & 0 & 0\\ -\frac{l_1\hat{J}_m}{\hat{K}_t} & -1 & 1 & 0\end{bmatrix} \tag{49}$$

$$A_s^{21} = \frac{-K_b\hat{K}_t - w_c\hat{L}_a l_1\hat{J}_m}{L_a\hat{K}_t} \tag{50}$$

$$B_s = \begin{bmatrix}0 & \frac{w_c\hat{L}_a}{L_a} & l_1 & 1\end{bmatrix}^T \tag{51}$$

$$C_s = \begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix} \tag{52}$$

$$x_d = \begin{bmatrix}\omega_m & i_a & \xi_1 & \xi_2 & \eta_2\end{bmatrix}^T \tag{53}$$

$$A_d = \begin{bmatrix} -\frac{B_m}{J_m} & \frac{K_t}{J_m} & 0 & 0 & 0\\ A_d^{21} & A_d^{22} & \frac{w_c\hat{L}_a}{L_a} & \frac{1}{L_a} & \frac{w_c\hat{R}_a}{L_a}\\ -\frac{l_1\hat{B}_m}{\hat{K}_t} & 0 & 0 & 0 & 0\\ A_d^{41} & A_d^{42} & l_2 w_c\hat{L}_a & 0 & l_2 w_c\hat{R}_a\\ -\frac{l_1\hat{J}_m}{\hat{K}_t} & -1 & 1 & 0 & 0\end{bmatrix} \tag{54}$$

TABLE V
SYSTEM PARAMETERS FOR G_i

	x	A	B	C
Conventional	x_{cc}	A_{cc}	B_{cc}	C_{cc}
PIO2:O	x_{dc}	A_{dc}	B_{dc}	C_{dc}

$$A_d^{21} = A_s^{21},\ A_d^{22} = \frac{-R_a - (l_2 + w_c)\hat{L}_a}{L_a} \tag{55}$$

$$A_d^{41} = \frac{-l_2 w_c \hat{L}_a l_1 \hat{J}_m}{\hat{K}_t}, \quad A_d^{42} = -l_2(\hat{R}_a + w_c \hat{L}_a) \tag{56}$$

$$B_d = \begin{bmatrix} 0 & \frac{w_c \hat{L}_a}{L_a} & l_1 & l_2 w_c \hat{L}_a & 1 \end{bmatrix}^T \tag{57}$$

$$C_d = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{58}$$

APPENDIX B
TRANSFER FUNCTIONS G_i

Similar to the case of G_ω, the transfer functions G_i from i_a^* to i_a is computed by (44) where the states and matrices are summarized in Table V.

The system parameters in Table V are given by the following matrices:

$$x_{cc} = x_c, \quad A_{cc} = A_c, \quad B_{cc} = B_c \tag{59}$$

$$C_{cc} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \tag{60}$$

$$x_{dc} = \begin{bmatrix} \omega_m & i_a & \xi_2 & \eta_2 \end{bmatrix}^T \tag{61}$$

$$A_{dc} = \begin{bmatrix} -\frac{B_m}{J_m} & \frac{K_t}{J_m} & 0 & 0 \\ -\frac{K_b}{L_a} & A_d^{22} & \frac{1}{L_a} & w_c \hat{R}_a L_a \\ 0 & A_d^{42} & 0 & l_2 w_c \hat{R}_a \\ 0 & -1 & 0 & 0 \end{bmatrix} \tag{62}$$

$$B_{dc} = \begin{bmatrix} 0 & \frac{w_c \hat{L}_a}{L_a} & l_2 w_c \hat{L}_a & 1 \end{bmatrix}^T \tag{63}$$

$$C_{dc} = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}. \tag{64}$$

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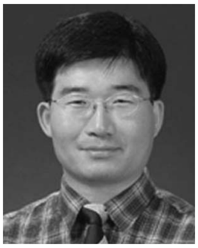

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